# Sanjana Kiran Urs

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## Professional Summary

**ML focused Software Engineer** with a graduate degree in Computer Science and hands-on expertise spanning end-to-end ML workflows: from ETL & statistical modeling to NLP and cloud-based inference. Passionate about applying strong foundations in statistics, Python, and SQL to solve real-world problems.

## Experience

**Research Intern** — Remote, USA
*Artificial Intelligence Innovation Laboratory - University at Buffalo* — June 2026 - Present
**Faculty Portal** | *MySQL, Java Spring Boot + React/Next.js, Project Management*

- Architecting a full-stack faculty portal end-to-end, serving **7+ REST APIs across 6 domains.** Working on improving data consistency, schema structure, and API performance.
- Redesigned a **14-table normalized schema** by eliminating redundant joins and cutting **average query response time by 40% across 8+ workflows** while **reducing peak DB read load by 35%.**

**Visiting Graduate Student** — Los Angeles, California, USA
*Cedars Sinai Medical Center - Slomka Lab* — July 2025 - August 2025

- Developed an interpretable **XGBoost** model to detect coronary artery disease in **3 weeks** using PET-derived imaging biomarkers and clinical variables from 11 multicenter cohorts. Implemented **LOSO cross-validation** (train on 10 sites, test on 1) to ensure cross-site generalizability, improving **AUC from 0.81 to 0.84.**
- **Accepted for Poster Presentation at American College of Cardiology (ACC)**

## Projects

**Emotion Detection in Social Media Posts** | *Python, PyTorch, HuggingFace, Scikit-learn, Gradio*

- Fine-tuned and compared three Transformer architectures **(RoBERTa, RoBERTa+CNN+NRC-lexicon, DeBERTa)** on Google's GoEmotions dataset with WordPiece tokenization and class weighting - achieving **85% test accuracy** and **reduced test loss from 0.68 to 0.11** with the enhanced model, with **DeBERTa yielding the highest F1 and ROC-AUC scores.**

**University Shop Management System** | *PostgreSQL, Indexing, Triggers, Python, NumPy, Pandas, Tableau*

- Designed a normalized PostgreSQL DB, implemented triggers, stored procedures, and complex queries to enforce data integrity and surface insights like top-rated shops and highest-spending users.
- Optimized query performance through strategic indexing and sub-query refactoring, achieving up to **10× reduction in execution cost** and enabling real-time Tableau dashboards for revenue and loyalty reporting.

**Your Personal AI Powered OS** | *FastAPI, Next.js, PostgreSQL, CI/CD, RESTfull API*

- Building a full-stack AI Personal Operating System, where users can manage tasks, goals, habits, notes, and calendar events, with an AI scheduling engine that auto-prioritizes tasks, detects conflicts, and replans missed work.

## Skills

**AI:** RAG, GenAI, Transformers, LLM fine tuning, Agentic workflows, Prompt Engineering
**Libraries:** NumPy, Pandas, SkLearn, Matplotlib, Scikit-learn, PyTorch, TensorFlow, Huggingface
**Databases & Big Data:** MySQL, PostgreSQL, Spark, Hadoop, Tableau, PowerBI, Excel, KPIs, Data Engineering
**Systems:** Data Structures, Algorithms, System Design, SDLC, CI/CD, Agile
**Tools and Frameworks:** JIRA, Postman, Linux, Docker, Git
**Data Science:** Mathematics, Regression, ETL, Advance analytics, Statistics, Data modeling, Data architecture AWS
**Languages:** Python, C, HTML, CSS, SQL, R, OOP in C++
**Soft Skills:** Problem-solving, Communication, High collaboration, Critical thinking, Decision making, CRM

## Education

**Master's degree in Computer Science** — New York, USA
*University at Buffalo - SUNY* | *GPA:* (3.7/4.0) — August 2024 - December 2025
**Coursework:** Algorithms, Machine Learning, Operating Systems, Deep Learning, Data Models & Query Language , Statistical Data Mining, Data Intensive Computing, Computer Vision, Intro to Pattern Recognition.

**Bachelor's degree in Information Science** — Mysore, India
*JSS Science and Technology University* | *GPA:* (3.87/4.0) — August 2019 - July 2023
**Coursework:** Artificial Intelligence, Data Mining, Database Management System, Object Oriented Programming in C++, Software Engineering, Linear Algebra, Data Structures & Algorithms, Operating Systems.